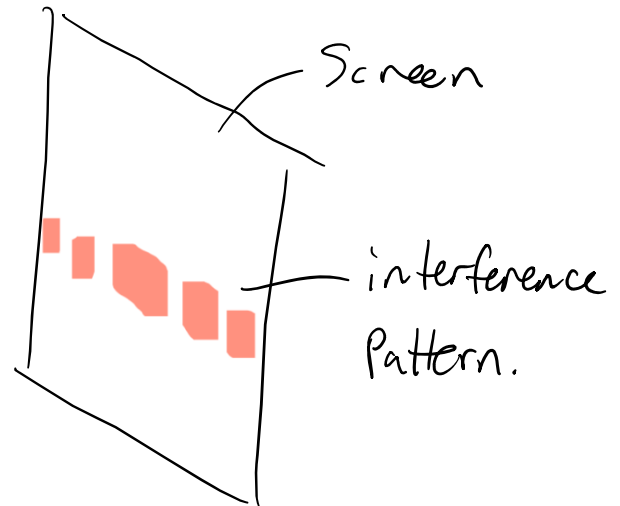
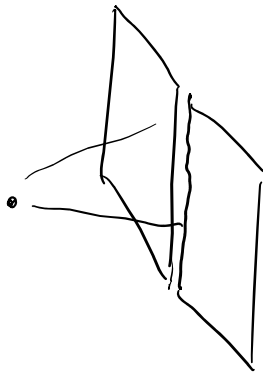
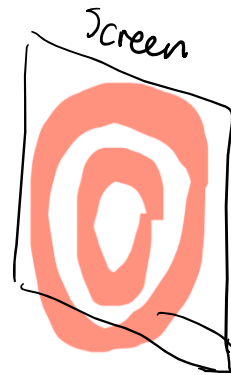
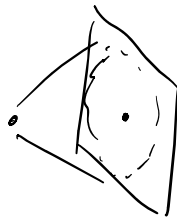


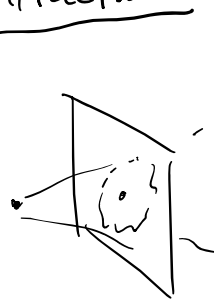
Diffraction



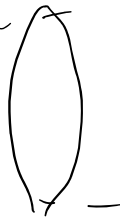
Fresnel Diffractions



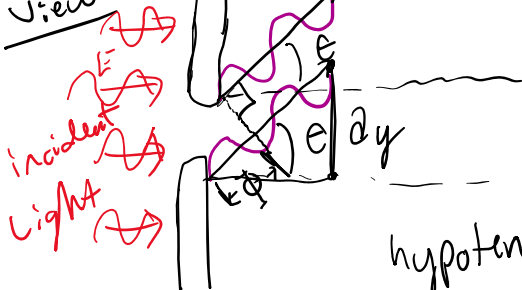
Fraunhofer Diffraction



lens

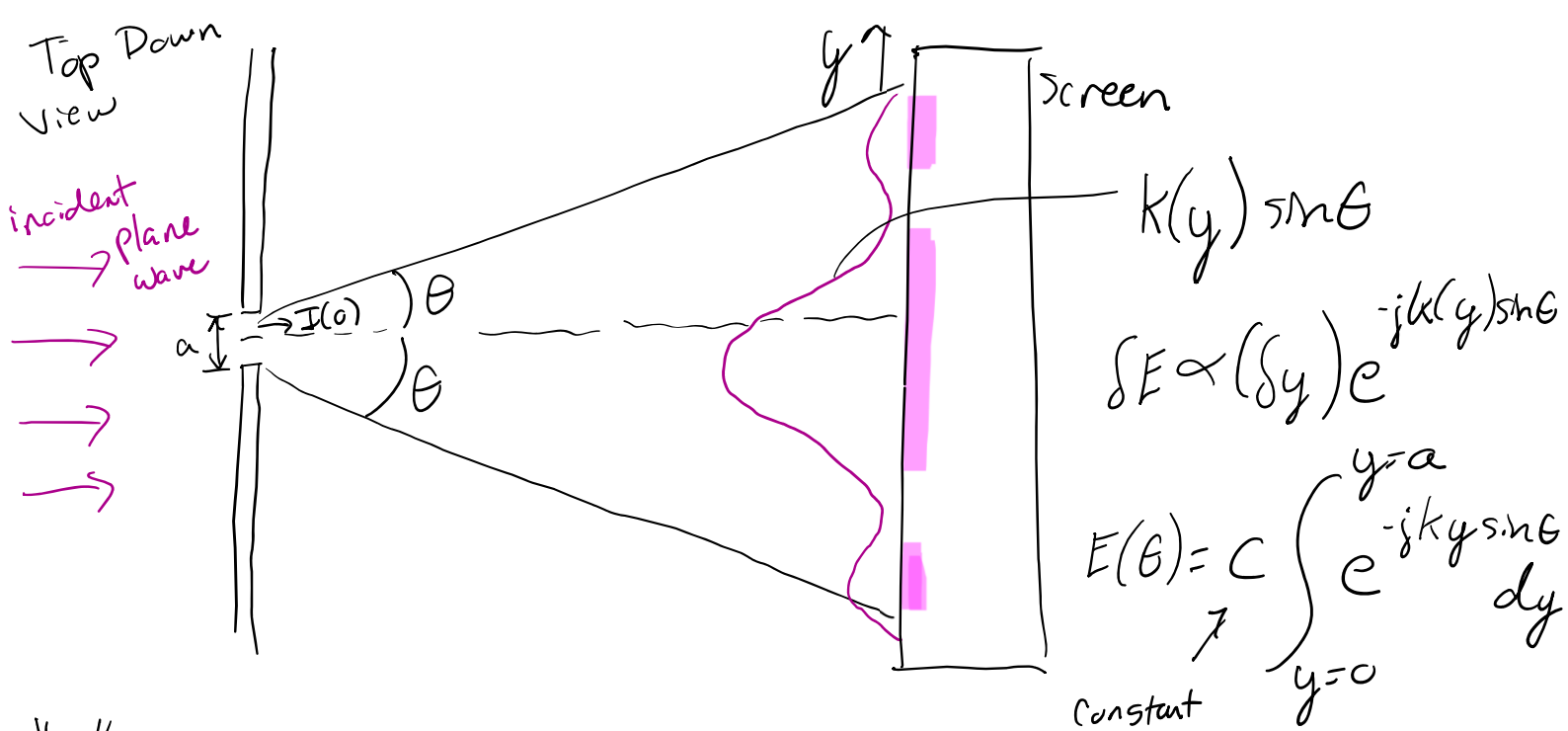


Top Down View



ϕ is proportional to

$$\text{hypotenuse} = dy \sin \theta$$



"a" aperture diameter

θ is the diffraction angle

$$E(\theta) = \frac{C^2 a^2 \left(e^{-j\frac{1}{2}k(a) \sin \theta} \right)^2 \left(\sin \left(\frac{1}{2}ka \sin \theta \right) \right)^2}{\left(\frac{1}{2}ka \sin \theta \right)^2}$$

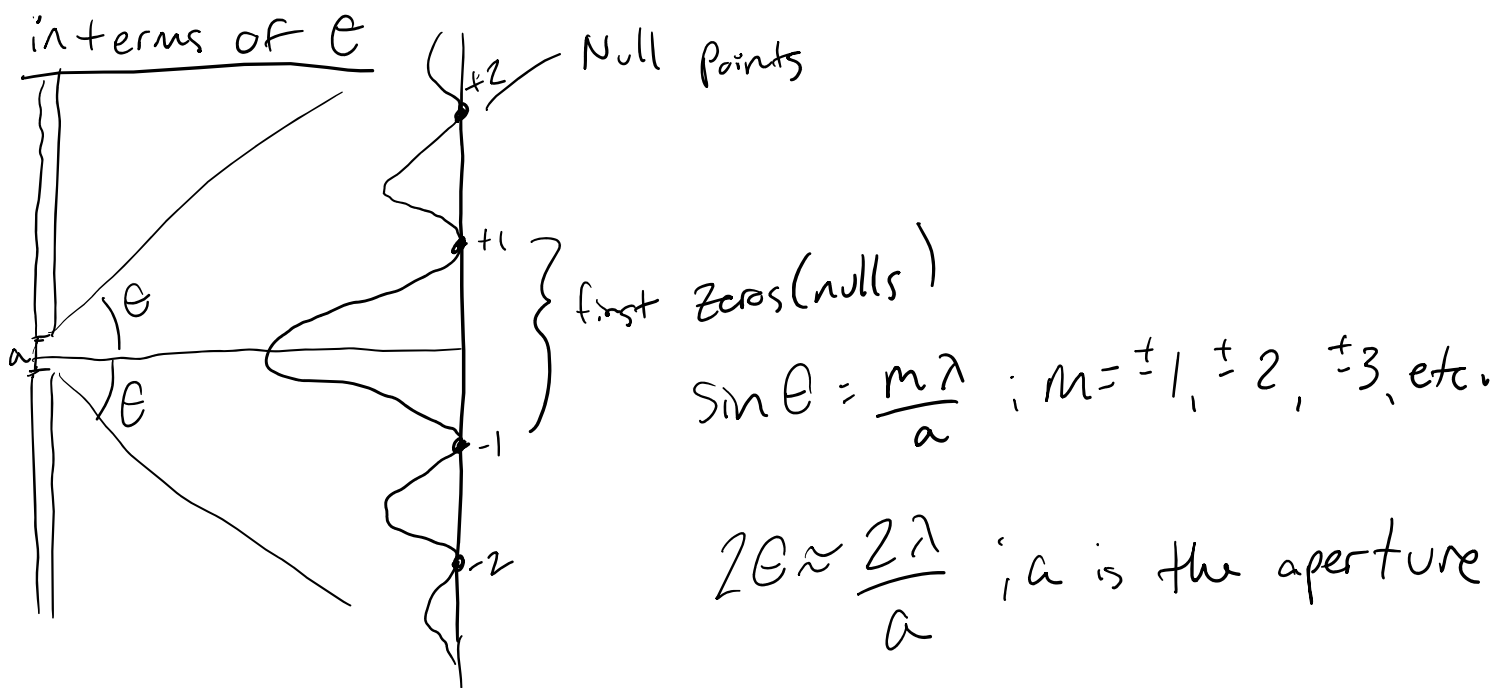
Intensity $(I(\theta)) \propto |E(\theta)|^2 = \left[\downarrow \right]^2$

$$E(\theta) = \frac{I(0) \sin \beta}{\beta} \Rightarrow \beta = \frac{1}{2}(ka \sin \theta)$$

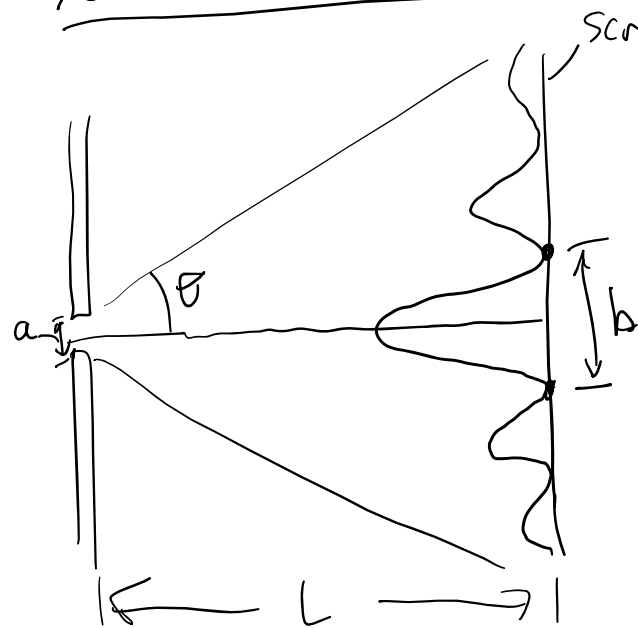
$$\boxed{I(\theta) = I(0) \text{sinc}^2 \beta}$$

$\frac{\sin \beta}{\beta} = \text{sinc} \beta$

$I(0)$ is the intensity normal to a ($\theta=0$)



Define in terms of the airy discs



$$b \ll L$$

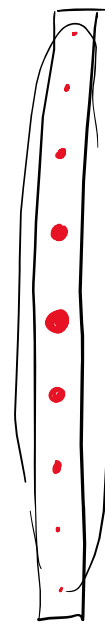
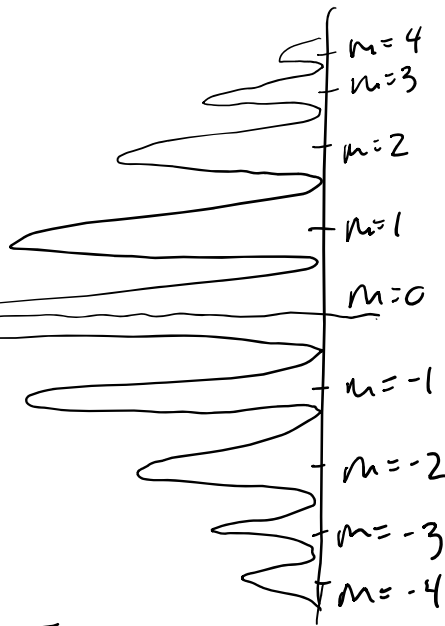
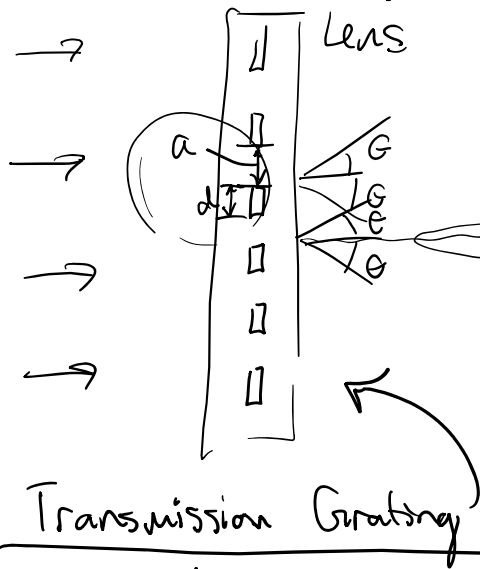
for an airy disc

$$\sin \theta = 1.22 \left(\frac{\lambda}{a} \right)$$

$$\frac{b}{2L} = \tan \theta \approx \theta$$

$$2\theta = \frac{b}{L}$$

Diffraction Gratings



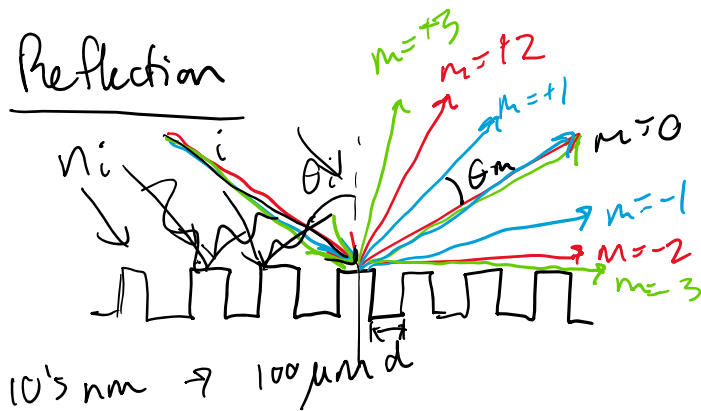
$$d \sin \theta = \frac{m \lambda}{n}$$

$$n = \sqrt{\epsilon_r}$$

- Bragg diffraction condition when the spacing of the slits (a) cause Φ delay that is equal to a full wavelength.

- most diffraction grates two types 1.) Reflection, 2.) Transmission

Reflection



$$d(n_i \sin \theta_i - n_r \sin \theta_m) = m \lambda \sin \phi$$

ϕ is the phase shift of the electric field.